

Portfolio

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High School: Sir Winston Churchill Secondary School, Vancouver, BC

Graduation Year: 2026

Program Applied: Faculty of Applied Science & Engineering
University of Toronto

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1. Personal Statement

My name is Thien Huynh, and I am going into Year 2 as an IB student at Sir Winston Churchill Secondary School in Vancouver, BC. Empathetic, collaborative and hands-on are qualities that I believe would apply to me and describes how I would approach things at both academic standpoints and the real world.

I left Vietnam to attend Perkiomen School in Pennsylvania on a full four-year scholarship at a young age of 13. Living and studying abroad, far away from my family or friends at such a young age taught me how to adapt and become more independent in an unknown environment, qualities that are fundamental to both my academic and personal development.

I have always been fascinated with how mechanical systems work. In my childhood, I had a phase of studying how bicycles work, how each component blend together to spin the wheel when you push on the pedal and how different types of bikes use different mechanics for their own respective purposes. This curious personality I carried with me until now, a passion to create. Thanks to this, a friend and I started our own project, *Bin Genius*, an AI-powered waste-sorting system that we made for the 2024 Youth Innovation Showcase and landed a position as a finalist in the competition, that was also recognized by CBC news. I have also taken on leadership roles in school-wide STEM initiatives and pursued internships that allow me to explore the intersection of design, engineering, and real-world impact.

Outside of academics, I am also an avid badminton player and have participated in many tournaments for my school team. The sport gave me an option to connect with my peers outside of the academic space and serves as a way to destress in academic heavy periods of my life.

I am drawn to the **University of Toronto's** Mechatronics Engineering for its unique and rigorous academic program that emphasizes a more hands-on approach to learning and that it offers an opportunity for me to both learn and work at the same time, allowing me to gain much valuable experience. I am eager to contribute **to University of Toronto's** learning environment where collaboration and curiosity are traits that are highly regarded, to be part of a community committed to building technologies that make a difference.

2. Resume

Education

Sir Winston Churchill Secondary School, Vancouver, BC, Canada

International Baccalaureate (IB) Program – Year 2

Years Attended: 2024 – 2026 (Grade 11–12)

Expected Graduation: June 2026

Gladstone Secondary School, Vancouver, BC, Canada

Year Attended: 2023 – 2024 (Grade 10)

Perkiomen School, Pennsylvania, USA

Year Attended: 2022 – 2023 (Grade 9)

EMASI Van Phuc School, Ho Chi Minh City, Vietnam

Years Attended: 2020 – 2022 (Grades 7–8)

Vietnam Australia International School, Ho Chi Minh City, Vietnam

Years Attended: 2014 – 2020 (Grades 1–6)

Experience

Bin Genius Founder

Vancouver, BC | 2023 – Present

<https://www.bingenius.org>

Led the development of "Bin Genius," an AI & IoT waste-sorting system designed to improve waste-sorting behavior through computer vision and IoT technology. This initiative involved designing the mechanical system, calibrating sensors, and developing the app interface, leading to the project's recognition as a finalist at the 2024 Youth Innovation Showcase

Engineering Intern – Kanovate Consulting Services Ltd

<https://kanovate.ca>

Product Design & Development Intern

Kanovate.ca – Vancouver, BC | June 2024 – Present

- Contributed to the development of early-stage product prototypes
- Assisted with mechanical design and applied tolerancing principles
- Used OnShape and TinkerCAD to create and test 3D-printed models
- Collaborated with startup clients to refine MVP concepts

Podcast Interviewer & Producer Intern

Kanovate.ca – Vancouver, BC | August 2024 – Present

- Conducted interviews with entrepreneurs and innovators for a digital podcast series
- Researched guest backgrounds and prepared insightful, topic-specific questions
- Assisted with post-production editing and publishing using podcast tools
- Helped shape storytelling, tone, and flow for engaging episode formats
- Coordinated scheduling and communication with featured guests

Skills

- Technical: Python, YOLOv8, AI model training, FlutterFlow, 3D printing, OnShape, TinkerCAD
- Soft Skills: Public speaking, leadership, collaboration, project management

Awards

- 1st place – Vancouver High School Badminton doubles 2025
- Finalist – Youth Innovation Showcase 2024
- Gold Medal – Hong Kong International Science Olympiad (HKISMO) 2021
- Silver Medal – World Scholar's Cup (Tournament of Champions, Yale University) 2019

3. Academic Highlights

- Predicted Score: 37/45
- Top Courses: IB program - HL Math, HL Physics, HL Chemistry, SL History, SL English, SL Japanese
- Extended Essay in History – a 4000-word extensive essay that aims to debate multiple historian perspectives on a specific topic, being “What were the most significant factors leading to the German’s defeat at Operation Barbarossa in 1941?”

4. Portfolio - Projects

Project: Bin Genius – AI & IoT Waste Sorting System

I led the development of “Bin Genius,” a smart waste bin system designed to promote better recycling habits through real-time waste recognition. Using YOLOv8 AI trained on over 5,000 images, the system achieved 80% classification accuracy. It integrates a VOC sensor to detect organic waste and provides user feedback via LED and buzzer. I was responsible for CAD prototyping, sensor integration, and front-end app development using FlutterFlow. The project was selected as a finalist at the 2024 Youth Innovation Showcase and is now being further improved to work out all the kinks that were identified by the judges and 3rd party comments from peers and experts.

<https://www.bingenius.org>

5. Reflections

At 13, I left Vietnam and earned a four-year scholarship to Perkiomen School in Pennsylvania. Living alone taught me how to adapt quickly, manage challenges, and grow from discomfort. Later moving to Canada pushed me to keep evolving, I carried with me a drive to learn and to lead. These diversified experiences from multiple different environments gave me a variety of perspectives both on the academics, and the community that I reside in.

One of my proudest academic moments came when I earned Gold at the Hong Kong International Science Olympiad. Competing across physics, chemistry and biology, I strengthened not just my scientific reasoning, but my confidence under pressure. It reinforced my passion for inquiry and my potential as a future engineer.

At Kanovate, I turned theory into practice. I built 3D CAD models and worked with clients to create podcasts. This experience sharpened my technical precision, collaboration, and iterative thinking, skills that are critical to engineering design.

I also believe that impact begins with empathy. In Long An - Vietnam, I organized a book donation drive for hospitalized children. I managed logistics, led fundraising, and delivered over 200 books, using leadership not for recognition, but for good. In Vancouver-Canada, I volunteered 36+ hours at Lynn Canyon Ecology Centre. There, I helped kids learn through hikes, crafts, and nature games. This helps me with practicing my patience, engagement, and clear communication.

Through CHAMP, Metro Vancouver's youth sustainability program, I explored the impacts of daily living on the environment, and the lengths that the community goes to reduce it. I joined workshops and a retreat to learn planning, analysis, and leadership. Now, I apply what I learned to green initiatives at my school, turning knowledge into action.

Leading Bin Genius, my AI powered waste sorting project, brought it all together. I led development, oversaw training data and UI design, and pitched at the 2024 Youth Innovation Showcase. That experience taught me how to blend engineering, teamwork, and storytelling. Our recognition affirmed this: real innovation is both technically sound and socially aware.

Along the way, I've competed as a badminton champion and performed Vietnamese sand writing at the World Scholar's Cup at Yale University. These moments remind me that identity and leadership are not separate, they deepen each other. Whether in labs, forests, or auditoriums, I lead by listening, building, and sharing with heart.

6. Letters of Reference

Letter from Mr. Jack Zissler – Owner and Director at 195 Global

Letter from Kanovate Consulting Services Ltd.

Letter from Teacher from Churchill

7. Supplementary Materials

- Certificate of Completion: UBC Geering Up Summer Camp 2025
- Youth4Action Leadership 2024 - Collaborative High School Action and Mentorship Program (CHAMP) Certification

Volunteer at Lynn Canyon Ecology Centre – Summer 2023 and 2024